Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	**Aim: To understand the process of convolution over the image and apply over the classification problem**	
Experiment No: 4	**Date: 11/03/2026**	**Enrolment No: 92410133046**

Aim: To understand the process of convolution over the image and apply it to a classification problem

using CNN.

IDE: Google Collab

Theory:

Convolutional Neural Networks (CNN) are a type of deep learning model mainly used for image classification and recognition. They work by extracting important features from images using layers like convolution, pooling, and fully connected layers. CNNs provide high accuracy because they automatically learn features such as edges, shapes, and patterns from images.

Methodology:

1. Load the basic libraries and packages
2. Load the dataset
3. Analyse the dataset
4. Normalize the data
5. Pre-process the data
6. Visualize the Data
7. Write the CNN model function
8. Write the Cost Function
9. Write the Gradient Descent optimization algorithm
10. Apply the training over the dataset to minimize the loss
11. Observe the cost function vs iterations learning curve

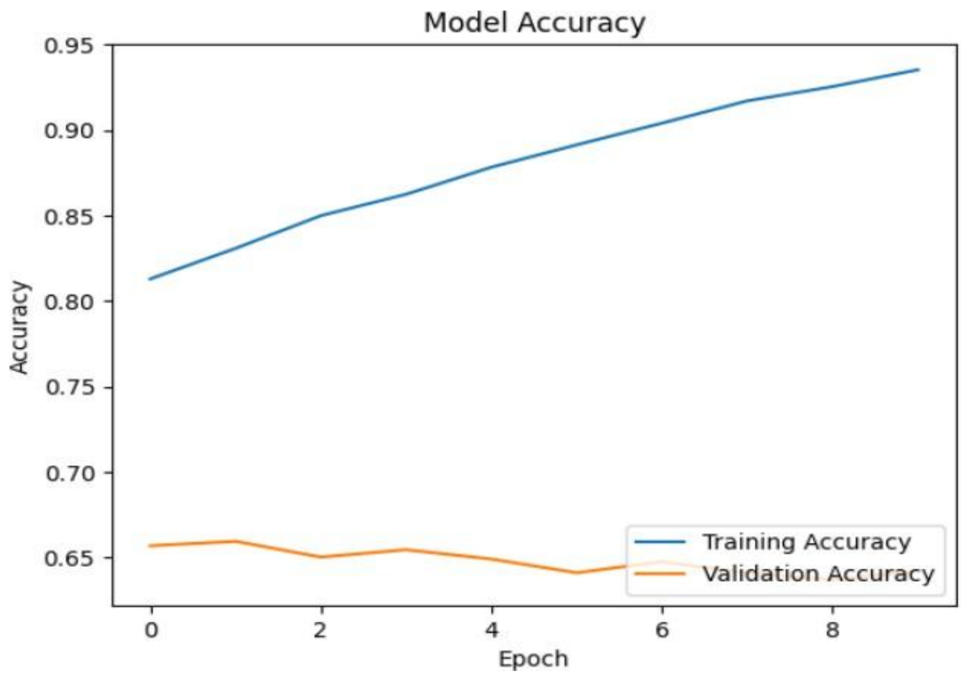
Program (Code):

```
import tensorflow as tf
from tensorflow.keras import layers, models

from tensorflow.keras.datasets import cifar10

from tensorflow.keras.utils import to_categorical

import matplotlib.pyplot as plt
```

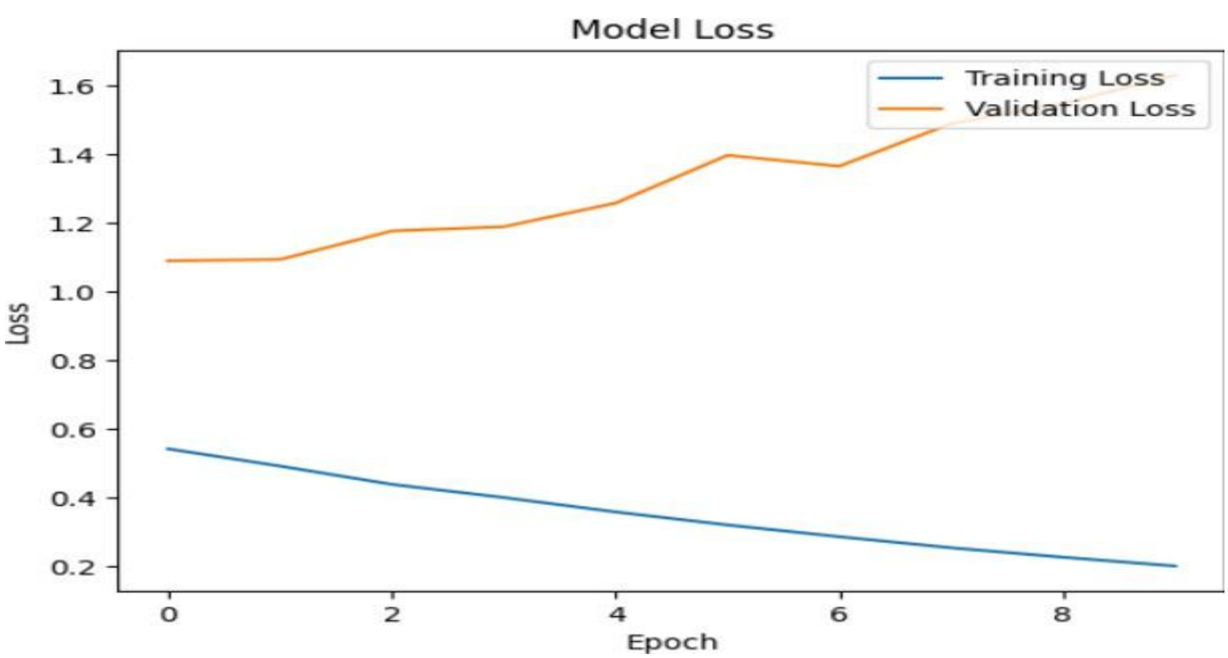
```
(train_images, train_labels), (test_images, test_labels) = cifar10.load_data()

train_images, test_images = train_images / 255.0, test_images / 255.0

train_labels = to_categorical(train_labels, 10)

test_labels = to_categorical(test_labels, 10)


model = models.Sequential([
    layers.Conv2D(32, (3, 3), activation='relu', input_shape=(32, 32, 3)),
    layers.MaxPooling2D(pool_size=(2, 2)),

    layers.Flatten(),

    layers.Dense(128, activation='relu'),

    layers.Dense(10, activation='softmax')

])


model.compile(optimizer='adam',

              loss='categorical_crossentropy',

              metrics=['accuracy'])


history = model.fit(train_images, train_labels, epochs=10, batch_size=64, validation_data=(test_images,
test_labels))


test_loss, test_acc = model.evaluate(test_images, test_labels)
```

```
print(f"Test accuracy: {test_acc * 100:.2f}%")


plt.plot(history.history['accuracy'], label='Training Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')

plt.title('Model Accuracy')

plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.legend(loc='lower right')

plt.show()

plt.plot(history.history['loss'], label='Training Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')

plt.title('Model Loss')

plt.xlabel('Epoch')

plt.ylabel('Loss')

plt.legend(loc='upper right')

plt.show()
```

Results:



Model Accuracy
Accuracy
Epoch
Training Accuracy
Validation Accuracy



Model Loss
Loss
Epoch
Training Loss
Validation Loss

Pre Lab Exercise:

a. **What is Convolution process? Explain giving example**.

Convolution is a mathematical operation used to combine two signals to produce an output that shows how one signal affects the other, commonly used in signal and image processing. It involves flipping one signal, shifting it over another, multiplying corresponding values, and summing them. For example, if $x[n] = \{1,2,3\}$and $h[n] = \{1,1\}$, their convolution gives $y[n] = \{1,3,5,3\}$, which represents the output signal.

b. **What are different layers in CNN model?**

The different layers in a CNN model are:

1. **Convolutional Layer** – extracts features from the input using filters
2. **Activation Layer (ReLU)** – adds non-linearity to the model
3. **Pooling Layer** – reduces the size of feature maps (downsampling)
4. **Fully Connected Layer** – connects neurons to make final predictions
5. **Dropout Layer** – prevents overfitting by randomly turning off neurons

c. **What is the requirement of the pooling layer in the CNN model?**

Pooling layer is used to reduce the size of feature maps. This helps in:
• Decreasing computational complexity
• Minimizing overfitting
• Retaining important features
• Max pooling selects the highest value, helping to remove noise

d. **What is the requirement of the use of ReLU activation function after convolution step?**

ReLU is used to eliminate negative values from the feature map by converting them to zero while keeping positive values unchanged. This helps in:

• Adding non-linearity to the model
• Making training faster
• Enhancing overall performance

<u>Observation and Result Analysis:</u>

e. **Nature of the dataset**

The dataset contains image data used for classification tasks. It includes labeled images that are preprocessed and normalized before training. It is divided into:
• Training set for learning the model
• Validation set for evaluating performance

f. **Training Process without regularization**

Without regularization:
• The model fits the training data very closely
• Training accuracy becomes high quickly
• Validation accuracy may remain low
• Overfitting occurs, where the model memorizes instead of generalizing

g. **After regularization in the training Process**

After applying regularization:
• Overfitting is reduced
• Training accuracy may slightly decrease
• Validation accuracy improves
• The model becomes more generalized and performs better on unseen data

h. **Observation over the Learning Curves**

• Training accuracy increases as the number of epochs increases
• Validation accuracy becomes stable after a certain point
• Loss decreases gradually during training
• Without regularization, there is a large gap between training and validation performance
• With regularization, both curves stay closer, showing better generalization

Post Lab Exercise:

a. **Why CNN is preferred over ANN for images**

CNN is preferred because:
- It automatically extracts important features from images without manual feature extraction
- It efficiently handles spatial data like images
- It uses fewer parameters compared to traditional ANN for image processing
- It provides higher accuracy in image classification and recognition tasks

b. **Can CNN be applied over Text data? If yes, then how. If no, then why?**

Yes, CNN can also be applied to text data.

How:
- Text is converted into numerical representations using word embeddings
- Convolution is applied over the embedded vectors to capture patterns like phrases or n-grams
- It is commonly used in tasks such as sentiment analysis and text classification

c. **What is the role of dropout layer?**

Dropout layer randomly turns off some neurons during training.

Purpose:
- Reduces overfitting
- Improves model generalization
- Makes the model more robust and less dependent on specific neurons

d. What will happen if maxpooling is replaced with minpooling?

If max pooling is replaced with min pooling:
- Important features may be ignored or lost
- Overall model performance may decrease
- More irrelevant noise may be included in the output
- Accuracy of the model will be lower

This is because max pooling selects the most significant (highest) values, while min pooling selects the least significant values from the feature map.

Conclusion:

The experiment showed how Convolutional Neural Networks (CNNs) are used for image classification. It helped in understanding the working of different layers such as convolution, pooling, and fully connected layers in extracting useful features from images and making predictions. The ReLU activation function improved the learning process by introducing non-linearity, while regularization techniques helped in reducing overfitting and improving generalization. Overall, the results proved that CNNs are a powerful and efficient model for image classification tasks.